

BedForge QC — Blueprint Intelligence

Blueprint Assessment Pack

Internal verification artifact. Compare side-by-side with plant shop drawings. Prestressed plant branding is secondary.

A. Cover / identity

Document ID	ffa2e566-def0-4d6b-8c19-b74d507dfaf4
Filename	L25390_Type2_I-Beam_Shop_Drawings_FLAT_BedForge.pdf
Uploaded	2026-08-19T16:10:50.705860+00:00
Updated	2026-08-19T21:32:35.681205+00:00
Page count	10
Linked job_id	—
Linked beam_id	—
Product type id	—
Upload hints	family=—; mark=—; project=—
Extraction status	NEEDS_REVIEW
Document status	needs_review
Locked revision	—
Extractor version	controlled_regex_ocr_v2
Extraction created	2026-08-19T21:32:35.676275+00:00
Confirmed / unconfirmed	20 / 30
Generated	2026-08-19 21:32 UTC

B. Critical identity

Large first-pass identity for plant-job comparison (Job L25390 / CID / Type 2 / marks / lengths). Schema-absent keys are listed explicitly so missing DNA is visible.

Job number (job_number)	L25390	high	confirmed	1	source=text_layer+ocr. Job number from JOB NO / L-number pattern.
CID (cid)	25-5390	high	confirmed	1	source=text_layer+ocr. CID from title block.
Beam mark (compact) (beam_mark)	201-209	high	confirmed	1	source=text_layer+ocr. Multi-beam identity compacted from beam_marks.
Beam marks (all) (beam_marks)	["201", "202", "203", "204", "205", "206", "207", "208", "209"]	high	confirmed	1	source=text_layer+ocr. Beam marks expanded from MARK/MK groups and ranges.
Product family / beam type (product_family)	i_beam	high	confirmed	1	source=text_layer+ocr. Type 2 / I-beam / FIB family callout.
County / DOT (county_dot)	Larue County, Kentucky	high	confirmed	1	source=text_layer+ocr. County/state identity from title block.

Bridge ID (bridge_id)	062B00034R	high	confirmed	1	source=text_layer+ocr. Bridge ID from title block.
Route (route)	KY 210	high	confirmed	1	source=text_layer+ocr. Route identifier from title block.
Overall length (overall_length_ft)	52.0	high	confirmed	7	source=text_layer+ocr. Primary overall length is the longest mark family; see mark_length_families for all groups.
Casting length (casting_length_ft)	52.0625	medium	confirmed	7	source=text_layer+ocr. Primary casting length is overall + 3/4" where the drawing says dimensions are BASED ON CASTING LENGTH.
Mark length families (mark_length_families)	[[{"marks": ["201", "202", "203"], "overall_display": "47'-3\"", "casting_display": "47'-3 3/4\"", "overall_length_ft": 47.25, "casting_length_ft": 47.3125, "source_page": 7, "casting_inferred_three_quarter": true}, {"marks": ["204", "205", "206"], "overall_display": "52'-0\"", "casting_display": "52'-0 3/4\"", "overall_length_ft": 52.0, "casting_length_ft": 52.0625, "source_page": 8, "casting_infe...	high	confirmed	7	source=text_layer+ocr. Per-mark overall/casting families. Missing casting lengths filled as overall + 3/4" from BASED ON CASTING LENGTH evidence.
Depth (overall_depth_in)	36.0	medium	confirmed	1	source=text_layer+ocr. AASHTO Type II / Type 2 I-beam overall depth is 36 in from section family callout (no explicit OVERALL DEPTH).
Strand diameter (strand_diameter_in)	0.5	high	confirmed	4	source=text_layer+ocr. Strand diameter from 1/2" or 0.5" callout.
Strand grade (strand_grade)	270K	high	confirmed	4	source=text_layer+ocr. Strand grade / relaxation from 270K callout.
Strand final pull (strand_final_pull_lb)	—	low	unconfirmed	—	Strand final pull not confidently located.
Hold-down type (hold_down_type)	H-56-S	high	confirmed	5	source=text_layer+ocr. Hold-down type from hardware callout.
Lift-loop spec (lift_loop_spec)	0.6" triple lift loops, embed 2'-7"	high	confirmed	10	source=text_layer+ocr. Lift-loop spec from hardware callout. Count not invented.

C. Full field inventory

Every FIELD_GROUPS key currently known to Blueprint Intelligence, including empty and not-applicable fields.

C1. Title Block / Identity (title_block)

job_number	L25390	high	confirmed	1	source=text_layer+ocr. Job number from JOB NO / L-number pattern.
cid	25-5390	high	confirmed	1	source=text_layer+ocr. CID from title block.
bridge_id	062B00034R	high	confirmed	1	source=text_layer+ocr. Bridge ID from title block.
route	KY 210	high	confirmed	1	source=text_layer+ocr. Route identifier from title block.
project_name	ING REBAR)	high	confirmed	7	source=text_layer+ocr. Project name from labeled field.
county_dot	Larue County, Kentucky	high	confirmed	1	source=text_layer+ocr. County/state identity from title block.
sheet_number	—	low	unconfirmed	—	Sheet number not confidently located.
revision	e	high	confirmed	4	source=text_layer+ocr. Revision from labeled field.
beam_mark	201-209	high	confirmed	1	source=text_layer+ocr. Multi-beam identity compacted from beam_marks.
beam_marks	["201", "202", "203", "204", "205", "206", "207", "208", "209"]	high	confirmed	1	source=text_layer+ocr. Beam marks expanded from MARK/MK groups and ranges.
product_family	i_beam	high	confirmed	1	source=text_layer+ocr. Type 2 / I-beam / FIB family callout.
overall_length_ft	52.0	high	confirmed	7	source=text_layer+ocr. Primary overall length is the longest mark family; see mark_length_families for all groups.
casting_length_ft	52.0625	medium	confirmed	7	source=text_layer+ocr. Primary casting length is overall + 3/4" where the drawing says dimensions are BASED ON CASTING LENGTH.
mark_length_families	[{"marks": ["201", "202", "203"], "overall_display": "47'-3\"", "casting_display": "47'-3 3/4\"", "overall_length_ft": 47.25, "casting_length_ft": 47.3125, "source_page": 7, "casting_inferred_three_quarter": true}, {"marks": ["204", "205", "206"], "overall_display": "52'-0\"", "casting_display": "52'-0 3/4\"", "overall_length_ft": 52.0, "casting_length_ft": 52.0625, "source_page": 8, "casting_inferred_three_quarter": true}, {"marks": ["207", "208", "209"], "overall_display": "47'-3\"", "casting_..."}]	high	confirmed	7	source=text_layer+ocr. Per-mark overall/casting families. Missing casting lengths filled as overall + 3/4" from BASED ON CASTING LENGTH evidence.

design_camber_in	—	low	unconfirmed	—	Design camber note not clearly stated.
finish_notes	—	low	unconfirmed	—	Finish notes present in drawing text; verify manually.

C2. Geometry (geometry)

overall_depth_in	36.0	medium	confirmed	1	source=text_layer+ocr. AASHTO Type II / Type 2 I-beam overall depth is 36 in from section family callout (no explicit OVERALL DEPTH).
top_flange_width_in	—	low	unconfirmed	—	Top flange width not confidently located.
top_flange_thickness_in	—	low	unconfirmed	—	Top flange thickness not confidently located.
bottom_flange_width_in	—	low	unconfirmed	—	Bottom flange width not confidently located.
bottom_flange_thickness_in	—	low	unconfirmed	—	Bottom flange thickness not confidently located.
web_thickness_in	—	low	unconfirmed	—	Web thickness not confidently located.
outer_width_in	—	low	not_applicable	—	source=text_layer. Not required for I-beam translation unless the drawing explicitly calls it out.
outer_depth_in	—	low	not_applicable	—	source=text_layer. Not required for I-beam translation unless the drawing explicitly calls it out.
wall_thickness_in	—	low	not_applicable	—	source=text_layer. Not required for I-beam translation unless the drawing explicitly calls it out.
void_width_in	—	low	not_applicable	—	source=text_layer. Not required for I-beam translation unless the drawing explicitly calls it out.
void_depth_in	—	low	not_applicable	—	source=text_layer. Not required for I-beam translation unless the drawing explicitly calls it out.
special_end_geometry	—	low	unconfirmed	—	Special end geometry/end-detail notes present; verify manually.

C3. Strand System (strand_system)

strand_count	—	low	unconfirmed	—	Strand count not confidently located.
strand_diameter_in	0.5	high	confirmed	4	source=text_layer+ocr. Strand diameter from 1/2" or 0.5" callout.
strand_grade	270K	high	confirmed	4	source=text_layer+ocr. Strand grade / relaxation from 270K callout.
strand_final_pull_lb	—	low	unconfirmed	—	Strand final pull not confidently located.

strand_area_in2	—	low	unconfirmed	—	Strand area not confidently located.
strand_pattern_rows	—	low	unconfirmed	—	Value was not confidently located.
straight_strand_count	—	low	unconfirmed	—	Straight strand count not clearly stated.
draped_strand_count	—	low	unconfirmed	—	Draped strand count not clearly stated.
hold_downs	—	low	unconfirmed	—	No deterministic hold-down station callouts were found in extractable text.
hold_down_type	H-56-S	high	confirmed	5	source=text_layer+ocr. Hold-down type from hardware callout.
jacking_force_kip	—	low	unconfirmed	—	Jacking force note not confidently located.
target_elongation_in	—	low	unconfirmed	—	Target elongation note not confidently located.
debond_notes	—	low	unconfirmed	—	Debonded or shielded strand notes present; verify manually.

C4. Hardware (hardware)

lift_loops	—	low	unconfirmed	—	No deterministic lift loop station callouts were found in extractable text.
lift_loop_spec	0.6" triple lift loops, embed 2'-7"	high	confirmed	10	source=text_layer+ocr. Lift-loop spec from hardware callout. Count not invented.
inserts	[{"type": "F-64"}]	high	confirmed	10	source=text_layer+ocr. F-64 insert callout captured without inventing quantity.
tubes	—	low	unconfirmed	—	No deterministic tube station callouts were found in extractable text.
tie_rod_openings	—	low	unconfirmed	—	No deterministic tie-rod opening station callouts were found in extractable text.
drain_holes	—	low	unconfirmed	—	No deterministic drain hole station callouts were found in extractable text.
stirrups	—	low	unconfirmed	—	Stirrup spacing not confidently located.
plates_and_clips	Plate, clip, or bearing notes present; verify manually.	medium	unconfirmed	10	source=text_layer+ocr. Keyword hit only; verify exact wording against source.
grout_grooves	—	low	not_applicable	—	source=text_layer. Not required for I-beam translation unless the drawing explicitly calls it out.
bituminous_ends	—	low	unconfirmed	—	Bituminous or cut-off pocket treatment noted; verify manually.

C5. Ends, QC & Finishes (ends_and_qc)

marked_end_rule	{"label": "", "end": "start"}	medium	confirmed	8	source=text_layer+ocr. Marked-end rule from labeled note.
end_elevation_notes	—	low	unconfirmed	—	End elevation note present; verify manually.
paint_id_requirements	Paint or ID marking note present; verify manually.	medium	unconfirmed	1	source=text_layer+ocr. Keyword hit only; verify exact wording against source.
dimensional_tolerances	—	low	unconfirmed	—	Tolerance note present; verify manually.
special_inspection_notes	Special inspection note present; verify manually.	medium	unconfirmed	1	source=text_layer+ocr. Keyword hit only; verify exact wording against source.

D. Missing / weak fields

CRITICAL_FIELDS that are empty, low-confidence, or UNCONFIRMED. These block or weaken lock / twin use.

All CRITICAL_FIELDS currently have a value and are not unconfirmed/low.

Other empty / unconfirmed / low-confidence fields

- bituminous_ends [unconfirmed/low] page=— value=—
- bottom_flange_thickness_in [unconfirmed/low] page=— value=—
- bottom_flange_width_in [unconfirmed/low] page=— value=—
- debond_notes [unconfirmed/low] page=— value=—
- design_camber_in [unconfirmed/low] page=— value=—
- dimensional_tolerances [unconfirmed/low] page=— value=—
- drain_holes [unconfirmed/low] page=— value=—
- draped_strand_count [unconfirmed/low] page=— value=—
- end_elevation_notes [unconfirmed/low] page=— value=—
- finish_notes [unconfirmed/low] page=— value=—
- hold_downs [unconfirmed/low] page=— value=—
- jacking_force_kip [unconfirmed/low] page=— value=—
- lift_loops [unconfirmed/low] page=— value=—
- paint_id_requirements [unconfirmed/medium] page=1 value=Paint or ID marking note present; verify manually.
- plates_and_clips [unconfirmed/medium] page=10 value=Plate, clip, or bearing notes present; verify manually.
- sheet_number [unconfirmed/low] page=— value=—
- special_end_geometry [unconfirmed/low] page=— value=—
- special_inspection_notes [unconfirmed/medium] page=1 value=Special inspection note present; verify manually.
- stirrups [unconfirmed/low] page=— value=—
- straight_strand_count [unconfirmed/low] page=— value=—
- strand_area_in2 [unconfirmed/low] page=— value=—
- strand_count [unconfirmed/low] page=— value=—
- strand_final_pull_lb [unconfirmed/low] page=— value=—
- strand_pattern_rows [unconfirmed/low] page=— value=—
- target_elongation_in [unconfirmed/low] page=— value=—
- tie_rod_openings [unconfirmed/low] page=— value=—
- top_flange_thickness_in [unconfirmed/low] page=— value=—
- top_flange_width_in [unconfirmed/low] page=— value=—
- tubes [unconfirmed/low] page=— value=—
- web_thickness_in [unconfirmed/low] page=— value=—

Schema gaps vs L25390-class shop drawings (not extracted today)

- bed layout / bulkhead stations (still plant-setup, not twin DNA)
- invented hardware quantities (intentionally omitted unless BOM text is explicit)

E. Page text evidence appendix

Sparse/empty text pages are rasterized and OCR'd (pytesseract). Field notes label source as text_layer, ocr, or text_layer+ocr.

Page 1 · source=text_layer+ocr

Merged excerpt (1237 chars, source=text_layer+ocr):

PRESTRESS SERVICES INDUSTRIES LLC
Flattened Shop Drawings - BedForge Upload Pack
Job No: L25390 | CID: 25-5390
KY 210 over North Fork Nolin River - Larue County, Kentucky
Bridge ID: 062B00034R | Type 2 I-Beam | Contractor: JAVE, LLC
Beam Marks: 201-209 (9 beams)
Images enhanced: contrast, sharpness, crop, orientation for AI extraction
BedForge QC development corpus · Flattened photo captures of accepted shop drawings
Flattened Shop Drawings – BedForge Upload Pack
Job No: L25390 | CID: 25-5390
KY 210 over North Fork Nolin River – Larue County, Kentucky
Bridge ID: 062B00034R | Type 2 I-Beam | Contractor: JAVE, LLC
Beam Marks: 201-209 (9 beams)
Images enhanced: contrast, sharpness, crop, orientation for AI extraction
BedForge QC development corpus - Flattened photo captures of accepted shop drawings
PRESTRESS SERVICES INDUSTRIES LLC

Flattened Shop Drawings – BedForge Upload Pack

Job No: L25390

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CID: 25-5390

KY 210 over North Fork Nolin River – Larue County, Kentucky

Bridge ID: 062B00034R | Type 2 I-Beam | Contractor: JAVE, LLC

Beam Marks: 201-209 (9 beams)

Images enhanced: contrast, sharpness, crop, orientation for AI extraction

BedForge QC development corpus - Flattened photo captures of accepted shop drawings

Page 2 · source=text_layer+ocr

Merged excerpt (5283 chars, source=text_layer+ocr):

L25390 · Sheet 1 of 17 · Cover Sheet / General Notes / Drawing Index
Job L25390 · Type 2 I-Beam · Larue County KY
2/10
Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam
L25390 - Sheet 1 of 17 - Cover Sheet / General Notes / Drawing Index
Job L25390 - Type 2 I-Beam - Larue County KY
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Page 3 · source=text_layer+ocr

Merged excerpt (3113 chars, source=text_layer+ocr):
 L25390 · Sheet 2 of 17 · Erection Layout & Details
 Beams 201-209 · Spans 1-3 · Diaphragms
 3/10
 Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam
 L25390 - Sheet 2 of 17 - Erection Layout & Details
 Beams 201-209 - Spans 1-3 - Diaphragms
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Page 4 · source=text_layer+ocr

Merged excerpt (3817 chars, source=text_layer+ocr):

L25390 · Sheet 3 of 17 · Casting Bed Layout (Span 2) + Strand Pattern Pours 1 & 2

MK 204/205/206 · 270' Bed · 1/2" 270K strands

4/10

Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam

L25390 - Sheet 3 of 17 - Casting Bed Layout (Span 2) + Strand Pattern Pours 1 & 2

MK 204/205/206 - 270' Bed - 1/2" 270K strands

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Page 5 · source=text_layer+ocr

Merged excerpt (4485 chars, source=text_layer+ocr):

L25390 · Sheet 4 of 17 · Casting Bed Layout (Spans 1 & 3) + Strand Pattern Pours 3 & 4
 MK 201-209 bed positions · Hold-downs H-56-S
 5/10
 Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam
 L25390 - Sheet 4 of 17 - Casting Bed Layout (Spans 1 & 3) + Strand Pattern Pours 3 & 4
 MK 201-209 bed positions - Hold-downs H-56-S
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L25390 · Sheet 5 of 17 · Reinforcement Details / Bar Bending
 #401(E) #403 #301 #302 · ASTM A615 Gr 60

6/10
 Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam
 L25390 - Sheet 5 of 17 - Reinforcement Details / Bar Bending
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 L25390 - Sheet 5 of 17 - Reinforcement Details / Bar Bending
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Page 7 · source=text_layer+ocr

Merged excerpt (3200 chars, source=text_layer+ocr):
 L25390 · Sheet 6 of 17 · Beam Shop Drawing - Marks 201 / 202 / 203
 47'-3" overall · Qty 1/3/1
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 Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam
 L25390 - Sheet 6 of 17 - Beam Shop Drawing - Marks 201 / 202 / 203

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Merged excerpt (3347 chars, source=text_layer+ocr):
L25390 · Sheet 7 of 17 · Beam Shop Drawing - Marks 204 / 205 / 206
52'-0" overall · Qty 1/3/1
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Merged excerpt (3046 chars, source=text_layer+ocr):

L25390 · Sheet 8 of 17 · Beam Shop Drawing - Marks 207 / 208 / 209

47'-3" overall · Qty 1/3/1

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Flattened for Blueprint Intelligence · Job L25390 Type 2 I-Beam

L25390 - Sheet 8 of 17 - Beam Shop Drawing - Marks 207 / 208 / 209

47'-3" overall - Qty 1/3/1

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 L25390 - Sheet 8 of 17 - Beam Shop Drawing — Marks 207 / 208 / 209

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Page 10 · source=text_layer+ocr

Merged excerpt (3291 chars, source=text_layer+ocr):
L25390 · Sheet 9 of 17 · Hardware - Mark 201
Lift loops · F-64 inserts · H-56-S hold-downs
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L25390 - Sheet 9 of 17 - Hardware - Mark 201
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Sheet 9 of 17

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F. Twin DNA readiness

Fields that would currently drive a 3D twin after lock via `normalize_locked_blueprint()`. Draft extractions do not render a production twin until Verify & Lock.

Twin lock state	NOT LOCKED — twin stays draft/legacy until lock
Normalized length	52.0
Cross-section keys	["bottom_flange_thickness_in", "bottom_flange_width_in", "overall_depth_in", "top_flange_thickness_in", "top_flange_width_in", "web_thickness_in"]
Strand pattern rows	0
Lift loops / hold-downs	0 / 0
Locked revision id	—

Would drive twin today (extracted + confirmed/manually_confirmed)

- `product_family` — Selects I-beam vs box-beam cross-section family
- `overall_length_ft` — Twin length (`normalized_blueprint.length`)
- `overall_depth_in` — I-beam overall depth
- `inserts` — Insert stations
- `marked_end_rule` — Marked-end label
- `strand_diameter_in` — Tension reference diameter
- `strand_grade` — 270K / low-relaxation identity
- `hold_down_type` — Hold-down hardware type (H-56-S)
- `lift_loop_spec` — Lift-loop specification (no invented counts)
- `beam_marks` — Multi-beam DNA mark list
- `mark_length_families` — Per-mark overall/casting length families
- `casting_length_ft` — Casting length for bed setup

Required twin inputs still missing or unconfirmed

- `top_flange_width_in` — I-beam top flange width
- `top_flange_thickness_in` — I-beam top flange thickness
- `bottom_flange_width_in` — I-beam bottom flange width
- `bottom_flange_thickness_in` — I-beam bottom flange thickness
- `web_thickness_in` — I-beam web thickness
- `strand_pattern_rows` — Strand row DNA (else invented from counts)
- `strand_count` — Fallback strand count
- `straight_strand_count` — Fallback straight-row count
- `draped_strand_count` — Drape profile + draped row
- `hold_downs` — Hold-down stations / drape low points
- `lift_loops` — Lift-loop stations
- `tubes` — Tube stations
- `tie_rod_openings` — Tie-rod stations
- `drain_holes` — Drain stations
- `stirrups` — Stirrup spacing (regions not modeled)
- `bituminous_ends` — Bituminous / cutoff pockets
- `jacking_force_kip` — Tension reference jacking force
- `target_elongation_in` — Tension reference elongation
- `strand_area_in2` — Tension reference area
- `strand_final_pull_lb` — Final pull (lb) into tension reference

Normalized DNA snapshot (what lock would persist)

```
{"cross_section": {"overall_depth_in": 36.0, "top_flange_width_in": null, "top_flange_thickness_in": null, "bottom_flange_width_in": null, "bottom_flange_thickness_in": null, "web_thickness_in": null}, "length": 52.0, "marked_end": {"label": "", "end": "start"}, "bituminous_ends": [], "lift_loops": [], "inserts": [{"type": "F-64"}], "tubes": [], "tie_rod_openings": [], "drain_holes": [], "hold_downs": [], "grout_grooves": [], "beam_marks": ["201", "202", "203", "204", "205", "206", "207", "208", "209"], "mark_length_families": [{"marks": ["201", "202", "203"], "overall_display": "47'-3\"", "casting_display": "47'-3 3/4\"", "overall_length_ft": 47.25, "casting_length_ft": 47.3125, "source_page": 7, "casting_inferred_three_quarter": true}, {"marks": ["204", "205", "206"], "overall_display": "52'-0\"", "casting_display": "52'-0 3/4\"", "overall_length_ft": 52.0, "casting_length_ft": 52.0625, "source_page": 8, "casting_inferred_three_quarter": true}, {"marks": ["207", "208", "209"], "overall_display": "47'-3\"", "casting_display": "47'-3 3/4\"", "overall_length_ft": 47.25, "casting_length_ft": 47.3125, "source_page": 9, "casting_inferred_three_quarter": true}], "cid": "25-5390", "bridge_id": "062B00034R",
```

```
"route": "KY 210", "casting_length_ft": 52.0625, "hold_down_type": "H-56-S", "lift_loop_spec": "0.6\" triple lift loops, embed 2'-7\""", "strand_grade": "270K", "strand_final_pull_lb": null, "tension_reference": {"target_elongation_in": null, "jacking_force_kip": null, "strand_area_in2": null, "strand_diameter_in": 0.5, "strand_final_pull_lb": null, "strand_grade": "270K"}, "dimensions": {"overall_length_ft": 52.0, "casting_length_ft": 52.0625, "overall_depth_in": 36.0}}
```

BedForge Blueprint Assessment — for print verification only
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